Formatting Bias in French LLM Evaluation: Evidence from the Compar:IA Arena

[Authors]

Abstract

LLM evaluation arenas, where users compare model outputs side-by-side, have become a primary source of model rankings. However, these rankings may conflate content quality with superficial formatting preferences. We analyze the Compar:IA dataset—a French government-backed LLM arena with 145,347 cleaned battles across 90 models—to quantify the effect of markdown formatting (bold, lists, headers) on user preferences. Using a style-controlled Bradley-Terry model, we find that each standard deviation increase in bold formatting, list items, or headers independently increases a model's win probability by 16–19%. After controlling for these formatting features, rankings shift substantially: 78 of 90 models show statistically significant rating changes after Benjamini-Hochberg correction for multiple comparisons (bootstrap CIs, n=1,000, FDR < 0.05). Formatting-heavy models drop sharply (mistral-medium-3.1: -22 ranks, mistral-large-2512: -20), while some reasoning models rise modestly (mean +2.6 ranks). Despite these per-battle effects, the overall ranking correlation between standard and style-controlled ratings remains high (r=0.976), suggesting that formatting bias inflates some models' positions without dominating the signal. Our analysis is the first to apply style control to a non-English LLM arena and reveals that formatting preferences observed in English-language evaluations are equally present in a French-speaking population.

1 Introduction

The rise of LLM evaluation arenas—platforms where users interact with two anonymous models and select a preferred response—has established a new paradigm for model comparison. The LMSYS Chatbot Arena pioneered this approach (Chiang et al. 2024), and its Elo-based rankings are widely cited as measures of model quality. The methodology has since been adopted by multiple platforms, including Compar:IA, a French government-backed arena launched in October 2024 (Termignon et al. 2026).

A key concern with arena-based evaluation is the extent to which user preferences reflect genuine content quality versus superficial presentation. Zheng et al. (2023) first noted that LLM judges exhibit a preference for longer, more verbosely formatted outputs. The LMSYS team subsequently introduced "style control"—a methodology that decomposes win probability into model skill and formatting effects using a modified Bradley-Terry model (Li et al. 2024). Their analysis of English-language data found that controlling for response length, markdown formatting, and list usage modestly reshuffled rankings.

We apply style control analysis to the Compar:IA dataset, which provides a unique opportunity:

1. **Scale:** 145,347 cleaned battles across 90 models, spanning 16 months of organic user interactions.
2. **Language:** French-language evaluation, enabling the first cross-linguistic test of whether formatting preferences are culturally invariant.
3. **Multi-signal design:** Both explicit votes (141K) and per-message reactions (27K) provide independent signals that we can cross-validate.
4. **Arena mode metadata:** The platform tracks whether model pairs were randomly assigned, user-selected, or drawn from specialized pools (reasoning, small models, big-vs-small).
5. **Reasoning models:** The dataset includes reasoning models (o3-mini, deepseek-r1, qwq-32b) whose chain-of-thought outputs produce systematically different formatting profiles.

Our central question: *Does controlling for markdown formatting change which models are ranked highest?*

2 Data

2.1 The Compar:IA Platform

Compar:IA is an LLM evaluation arena operated by the French government's Direction interministérielle du numérique (DINUM). Users submit prompts and receive responses from two anonymous models side-by-side, then vote for a winner or declare a tie. The platform also supports per-message reactions (like/dislike) with optional quality attribute annotations. Models are identified only after voting.

The platform offers five arena modes: **random** (56%), **custom** (27%, user-selected model pairs), **big-vs-small** (5%, deliberately pairing large and small models), **reasoning** (3%, ensuring at least one reasoning model), and **small-models** (2%).

2.2 Datasets

We use three datasets published on HuggingFace (`ministere-culture/comparia-*`):

Dataset	Raw Rows	After Cleaning	Content
Conversations	459,849	—	Full response text, metadata
Votes	148,957	141,059	Explicit winner selections
Reactions	89,717	88,939	Per-message like/dislike

2.3 Data Quality and Cleaning

We applied the following filters, documented in a comprehensive data quality audit:

Votes. Removed 503 same-model pairs (0.34%), 3,994 no-choice votes (user revealed models without voting), and 3,401 duplicate conversation_pair_ids (keeping most recent).

Reactions. Removed 352 same-model pairs, 995 reactions on even-indexed messages (user messages rather than model responses), and 431 reactions on empty/very short responses (<10 characters).

Reaction-to-vote conversion. Reactions were converted to pairwise preferences: for each conversation with reactions on both models, we computed a net score (likes minus dislikes) per model and assigned the winner by higher net score, with ties when scores were equal. This produced 26,673 reaction-derived battles from 50,849 unique conversations; the remaining 45.4% of conversations had reactions on only one model and were excluded (requiring an inner merge).

Combined dataset (before reasoning filter). 167,724 battles (141,059 from explicit votes + 26,665 from reactions). Only 8 conversation_pair_ids appeared in both sources; these were deduplicated in favor of explicit votes.

Reasoning-only content filter. We identified 25,778 battles (15.1%) where at least one model's response had empty user-visible `content` but non-empty `reasoning` or `reasoning_content` fields—indicating that the chain-of-thought was recorded but the final response text was lost. These battles cannot be meaningfully analyzed for formatting bias since all style features are necessarily zero due to missing text, not because the model produced unformatted output. Removing these battles yielded a final dataset of **145,347 battles across 90 models** with ≥100 battles each. The most affected models were gemini-3-pro-preview (2,635 battles removed), gpt-5-mini (2,383), qwen3-30b-a3b (2,320), and qwen-3-8b (2,311).

2.4 Notable Data Quality Findings

Reasoning content separation. The dataset stores chain-of-thought reasoning in dedicated `reasoning` and `reasoning_content` fields for 27 reasoning-capable models. However, for a substantial fraction of these models' battles, the user-visible `content` field is empty while reasoning fields are populated—indicating a data pipeline issue where only the chain-of-thought was persisted. These 25,778 affected battles were excluded (see reasoning-only content filter above). Among the remaining battles, only 117 messages (0.08%) across 98 conversations had `<think>` tags leaked into the user-visible content field, primarily from `qwen2.5-coder-32b-instruct` (82) and `deepseek-r1-distill-llama-70b` (33). These were filtered during style feature extraction.

Security probes. 56 entries in the arena mode column contained SQL injection, XSS, and SSRF payloads—artifacts of penetration testing against the platform. These were mapped to the "unknown" mode category.

Reaction-derived tie inflation. The structural properties of binary reactions produce a 46.8% tie rate in reaction-derived battles versus 30.7% in explicit votes. When a user likes both models or dislikes both, the conversion always yields a tie. Reaction-derived battles therefore provide less discriminating information per battle.

Near-zero overlap. Only 8 conversation_pair_ids appear in both the votes and reactions datasets. The two signal sources come from almost entirely different conversations, making cross-validation meaningful.

3 Methodology

3.1 Style Features

We extracted five formatting features from each model's response text:

Feature	Description	Regex Pattern
Headers	Markdown headers (# through ######)	`^#{1,6}\s` (multiline)
Lists	Ordered and unordered list items	`^\s*[-*+]\s` and `^\s*\d+\.\s`
Bold	Bold-formatted text	`\*\*[^*]+\*\*`
Code blocks	Fenced code blocks	`"‘` or `~~~` (counted in pairs)
Emoji	Emoji characters	Unicode emoji ranges

We deliberately excluded response length from the style control. Length was the dominant feature in the original LMSYS style control analysis, but it confounds with completeness—a core quality dimension. Users in a French-language arena may genuinely prefer more thorough answers, and controlling for length risks removing legitimate quality signal. The `complete` quality attribute in reaction data correlates with higher like rates (28.9% of liked messages were tagged "complete" versus 0% of disliked), supporting the view that length partly proxies for quality rather than mere formatting preference.

3.2 Bradley-Terry Model

We rank models using a Bradley-Terry model (Bradley and Terry 1952) estimated via logistic regression, following the methodology of the LMSYS Chatbot Arena (Chiang et al. 2024).

Standard model. For each decisive battle (A wins or B wins), we construct a feature vector with $+1$ for the winning model's index and -1 for the losing model's index, then fit a logistic regression without regularization. Ratings are computed as:

$$\text{Rating}_i = 1000 + \frac{400 \cdot \beta_i}{\ln(10)} \tag{1}$$

Style-controlled model. We augment the model indicator features with style difference features. For each formatting feature f, we compute $\Delta f = f_A - f_B$ (standardized to zero mean and unit variance), then fit:

$$P(\text{A wins}) = \sigma\left(\sum_i \beta_i \cdot \mathbb{1}_i + \sum_f \gamma_f \cdot \Delta f\right) \tag{2}$$

where β_i are model skill parameters and γ_f are style coefficients. The style-controlled rating uses only the β_i coefficients, with formatting effects absorbed by γ_f.

3.3 Bootstrap Inference

We computed 95% confidence intervals via nonparametric bootstrap (1,000 iterations for both style coefficients and BT ratings). Each bootstrap sample drew battles with replacement from the full dataset and re-estimated the model.

For each test, we derived two-sided bootstrap p-values as $p = 2 \cdot \min(\hat{F}(0), 1 - \hat{F}(0))$, where $\hat{F}(0)$ is the proportion of bootstrap replicates with the statistic ≤ 0, with a floor of $1/(B+1)$ to avoid zero p-values.

3.4 Multiple Comparison Correction

We applied the Benjamini-Hochberg (BH) procedure (Benjamini and Hochberg 1995) to control the false discovery rate (FDR) at 0.05. Corrections were applied separately to two families of tests: the 5 style coefficient tests and the 90 model rank change tests. The BH procedure ranks p-values and adjusts each as $p_{\text{BH}}^{(i)} = p^{(i)} \cdot m/i$, enforcing monotonicity via step-up.

4 Results

4.1 Style Coefficients

Table 1 shows the estimated effect of each formatting feature on win probability, after controlling for model identity. Figure 1 presents these coefficients as a forest plot with bootstrap confidence intervals.

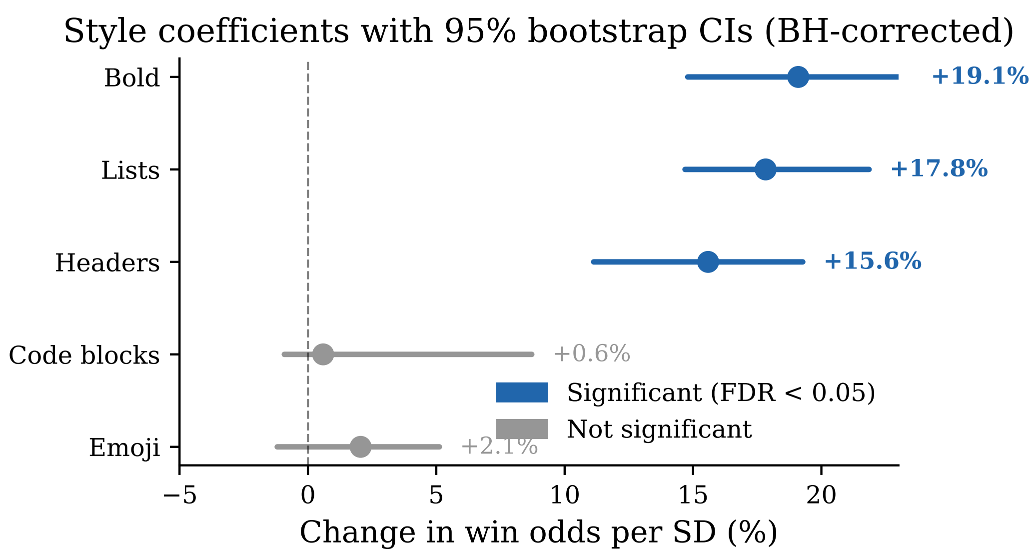

Figure 1: Style coefficients with 95% bootstrap CIs (BH-corrected). Bold, lists, and headers each increase win odds by 16–19% per SD, while code blocks and emoji have negligible effects.

Table 1: Style coefficients from the Bradley-Terry model (145,347 battles, 90 models). p-values are BH-adjusted across 5 tests.

Feature	Coeff.	95% CI	% Odds Δ	95% CI	p (BH)	Sig?
Bold	+0.175	[+0.138, +0.211]	+19.1%	[+14.8%, +23.5%]	0.003	Yes
Lists	+0.164	[+0.137, +0.198]	+17.8%	[+14.7%, +21.9%]	0.003	Yes
Headers	+0.145	[+0.106, +0.176]	+15.6%	[+11.1%, +19.3%]	0.003	Yes
Code blocks	+0.006	[−0.009, +0.084]	+0.6%	[−0.9%, +8.7%]	0.542	No
Emoji	+0.020	[−0.012, +0.050]	+2.1%	[−1.2%, +5.1%]	0.333	No

A one-standard-deviation increase in bold formatting gives a response a 19.1% boost in win odds, independent of which model produced it. Lists and headers have comparable effects (+17.8% and +15.6%). Code blocks and emoji have negligible effects that are not statistically distinguishable from zero. These effects are somewhat larger than before the reasoning-only content filter, consistent with the filter having removed battles where zero-valued style features diluted the estimated formatting effect.

The three significant features—bold, lists, and headers—all relate to structural formatting that makes responses visually organized. The null effect for code blocks suggests that code formatting per se does not influence preferences (its effect may depend on the task being coding-related). The null effect for emoji runs counter to the popular assumption that emoji-laden responses are preferred.

4.2 Ablation Study

To understand which features contribute most to rank changes, we ran the BT model controlling for one feature at a time (Table 2, Figure 2).

Table 2: Ablation: single-feature style control.

Feature Controlled	Coefficient (alone)	Rank Corr. with Standard
Bold only	+0.310 (+36.3%)	0.978
Lists only	+0.278 (+32.0%)	0.993
Headers only	+0.256 (+29.1%)	0.993
Code blocks	+0.068 (+7.0%)	1.000
Emoji only	+0.102 (+10.7%)	0.998
All five	**(see Table 1)**	**0.976**

When features are controlled individually, their coefficients are roughly double those in the joint model (e.g., bold alone: +36.3% vs. bold joint: +19.1%). This reflects the correlation among formatting features—models that use more bold also tend to use more lists and headers. The joint model partitions the shared variance among correlated features. Bold alone produces the most rank disruption (r=0.978), consistent with it having the largest joint coefficient.

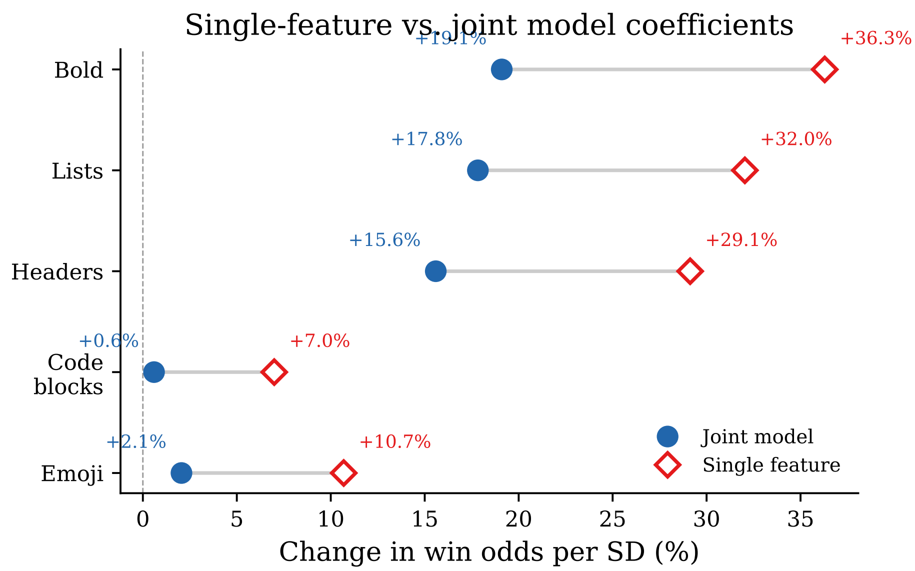

Figure 2: Single-feature vs. joint model coefficients. The gap shows the shared variance among correlated formatting features.

4.3 Ranking Impact

The overall correlation between standard and style-controlled BT ratings is $r = 0.976$ (Figure 3). Rankings are highly stable overall, but specific models shift substantially.

Table 3: Top 10 models: standard vs. style-controlled rankings. Bold entries in the standard column are absent from the style-controlled top 10.

Standard Ranking			**Style-Controlled Ranking**		
Rank	Model	Rating	Rank	Model	Rating
1	gemini-3-pro-preview	1288.0	1	gemini-3-pro-preview	1285.0
2	gemini-3-flash-preview	1201.3	2	gemini-3-flash-preview	1196.1
3	**mistral-large-2512**	1176.9	3	gemini-2.5-flash *(std #5)*	1137.9
4	**mistral-medium-2508**	1169.7	4	magistral-medium *(std #6)*	1128.8
5	gemini-2.5-flash	1166.8	5	gemini-2.0-flash *(std #9)*	1121.0
6	magistral-medium	1147.9	6	qwen3-max *(std #7)*	1119.9
7	qwen3-max-2025-09-23	1145.1	7	kimi-k2-thinking *(std #21)*	1108.2
8	**mistral-medium-3.1**	1131.7	8	grok-4.1-fast *(std #18)*	1105.7
9	gemini-2.0-flash	1129.7	9	mistral-medium-2508 *(std #4)*	1105.6
10	gpt-5.1	1126.2	10	gpt-5.1 *(std #10)*	1103.7

Three Mistral models that dominate standard rankings (#3, #4, #8, bolded) drop out of or to the bottom of the style-controlled top 10—replaced by kimi-k2-thinking and grok-4.1-fast, models that produce less formatted output. Figure 4 shows the top 20 rank movers.

7 of 10 top rank changes remain statistically significant after BH correction (FDR < 0.05). Across all 90 models, **78 of 90 (87%)** show significant rating changes after style control, indicating that formatting bias affects the vast majority of models, not just a few outliers.

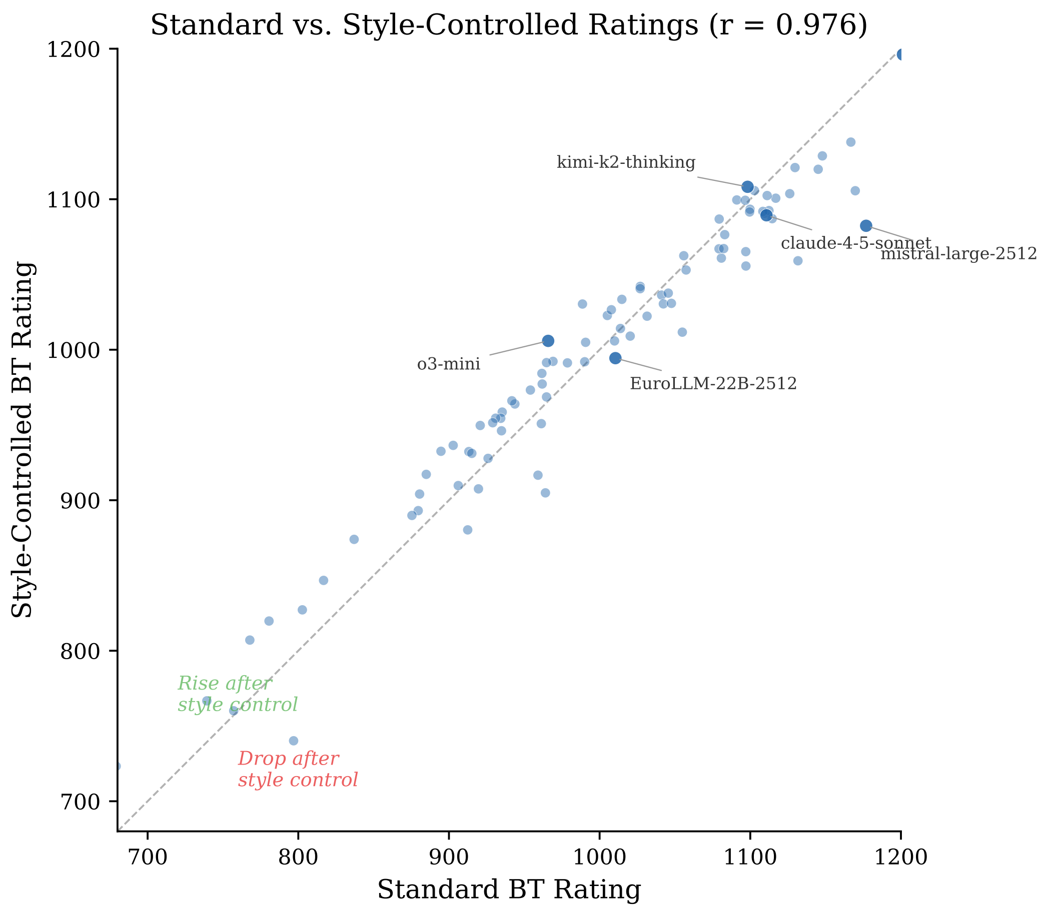

Figure 3: Standard vs. style-controlled BT ratings for all 90 models ($r = 0.976$). Models above the diagonal rise after style control; those below drop.

The most dramatic shifts are in the Mistral model family: **mistral-medium-3.1** drops from rank 8 to rank 30 (−72.6 rating points) and **mistral-large-2512** drops from rank 3 to rank 23 (−94.6 rating points). These models produce heavily formatted outputs (abundant bold, headers, and lists) that are rewarded in standard rankings but absorbed by the style control.

4.4 Reasoning Models and Formatting

Reasoning models show a mixed but on-average positive pattern after style control.

Mean rank change for reasoning models: +2.6 (modestly rising). Four of nine rise, one holds position, and four fall—a pattern that is directionally positive but less uniform than might be expected. The two largest risers—kimi-k2-thinking (+14) and grok-3-mini-beta (+12)—are not statistically significant after BH correction, likely because the reasoning-only content filter removed many of their battles, reducing statistical power.

This mixed pattern has an important methodological implication. Before applying the reasoning-only content filter (Section 2.3), the mean reasoning model rank change was +4.4 with 7 of 9 models

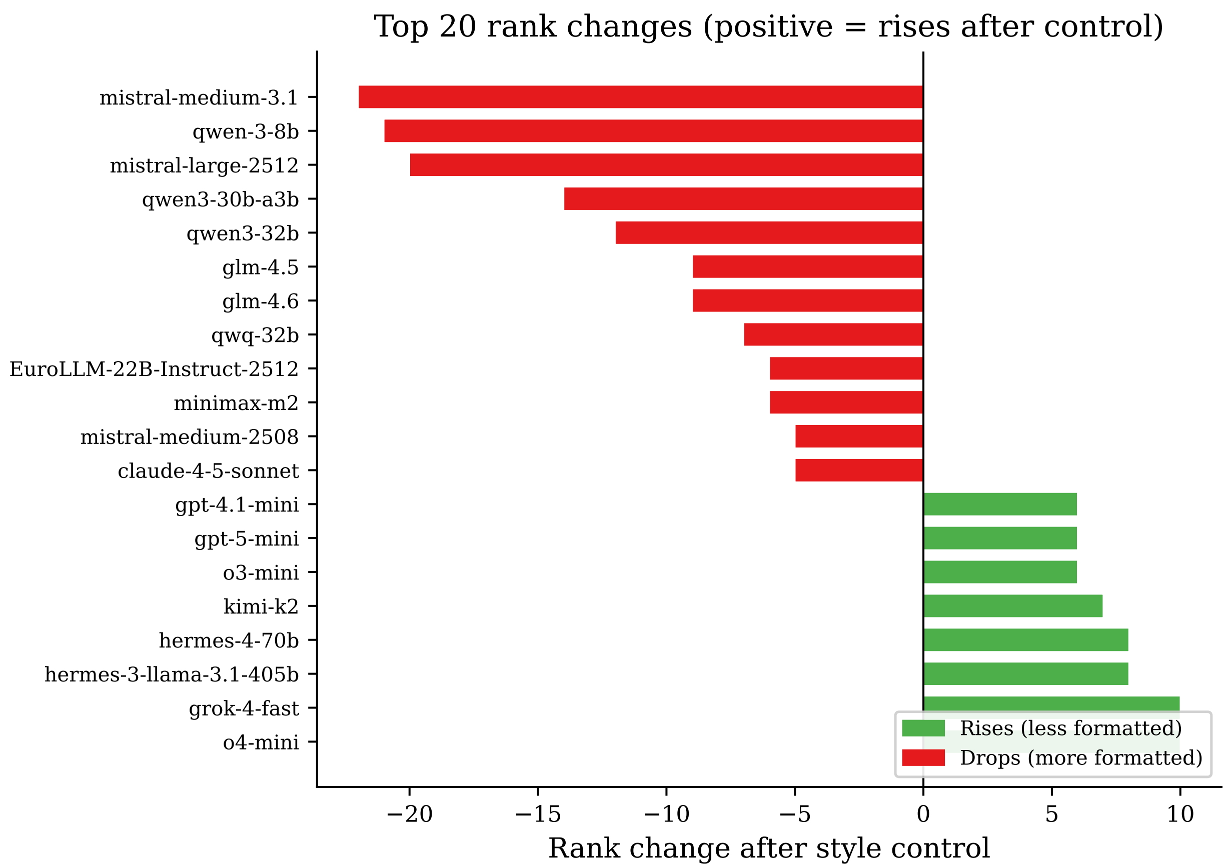

Figure 4: Top 20 rank changes after style control. Hatched bars indicate changes not significant after BH correction.

rising or holding. The reduction to +2.6 after the filter demonstrates that the earlier finding was partially inflated by battles where reasoning models had zero-valued style features due to missing content rather than genuinely plain-text output. The corrected pattern suggests that while some reasoning models do benefit from style control (plausibly because they prioritize reasoning depth over visual formatting), others produce output that is comparably or even more formatted than non-reasoning models.

4.5 Position Bias

We observe a statistically significant but negligible position bias: model A wins 50.40% of decisive battles versus 49.60% for model B (binomial test $p = 0.013$). The effect size (0.8 percentage points) is too small to meaningfully affect rankings.

4.6 Sensitivity Analyses

Votes only versus combined. BT ratings computed from explicit votes alone correlate at $r = 0.940$ with the combined (votes + reactions) ratings, with a maximum rank difference of 46

Table 4: Top 10 rank changes after style control. p-values are BH-adjusted across all 90 models (FDR < 0.05).

Model	Std Rank	Ctrl Rank	ΔRank	ΔRating	p (BH)	Sig?
mistral-medium-3.1	8	30	**−22**	−72.6	0.003	Yes
qwen-3-8b	56	77	**−21**	−59.1	0.003	Yes
mistral-large-2512	3	23	**−20**	−94.6	0.003	Yes
qwen3-30b-a3b	60	74	**−14**	−42.5	0.003	Yes
kimi-k2-thinking	21	7	+14	+10.0	0.572	No
qwen3-32b	33	45	**−12**	−43.2	0.003	Yes
grok-3-mini-beta	25	13	+12	+8.5	0.358	No
grok-4-fast	24	14	**+10**	+2.6	0.012	Yes
o4-mini	50	40	**+10**	+41.7	0.003	Yes
grok-4.1-fast	18	8	+10	+3.0	0.358	No

Table 5: Reasoning model rank changes. †Significant by rating change, not rank change.

Model	Std → Ctrl Rank	ΔRank	Sig?
kimi-k2-thinking	21 → 7	+14	No
grok-3-mini-beta	25 → 13	+12	No
o4-mini	50 → 40	**+10**	Yes
o3-mini	53 → 47	**+6**	Yes
deepseek-r1-distill-llama-70b	75 → 75	0	Yes†
deepseek-r1-0528	13 → 16	**−3**	Yes
olmo-3-32b-think	85 → 89	−4	No
deepseek-r1	38 → 43	**−5**	Yes
qwq-32b	74 → 81	**−7**	Yes

positions. The reasoning-only content filter disproportionately removed vote battles (22,521 of 25,778 removed), which reduced the vote-only sample to 118,538 battles and increased the relative weight of reaction-derived battles in the combined rankings. This explains the lower correlation compared to what would be expected with the full vote dataset.

Reaction versus vote agreement. Among the 85 models with sufficient battles in both sources, reaction-derived BT ratings correlate at $r = 0.867$ with vote-derived ratings (Figure 5). This agreement from two independent signal sources—with near-zero conversation overlap—supports the validity of both measures, though the moderate correlation also indicates meaningful disagreement on some models' relative positions.

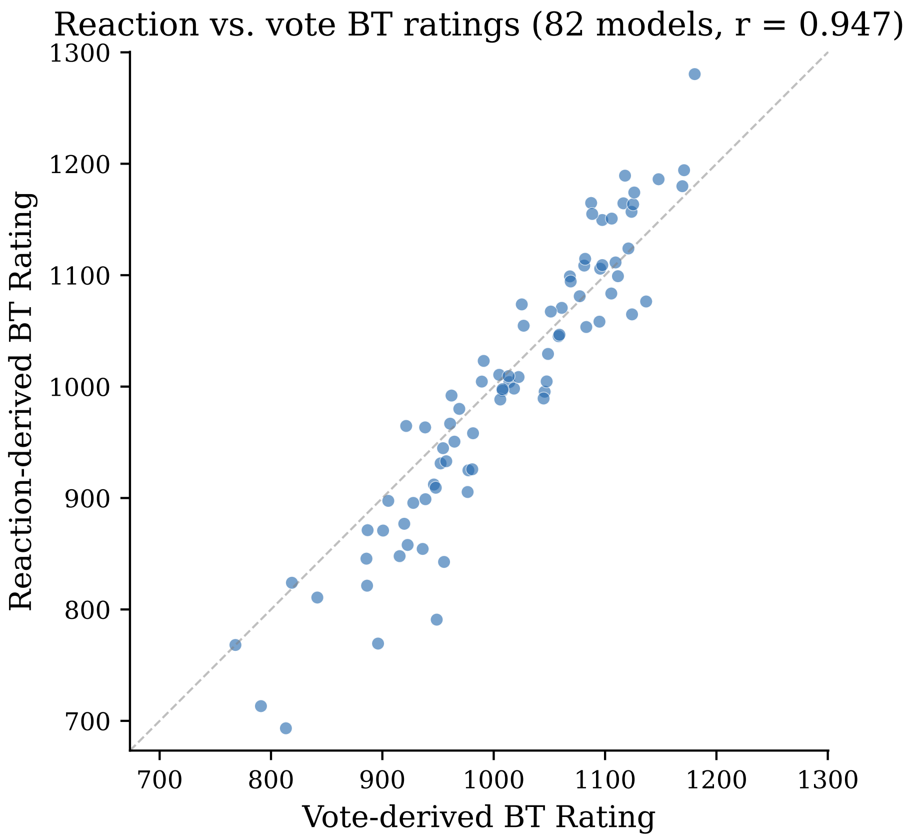

Figure 5: Reaction-derived vs. vote-derived BT ratings. Agreement ($r = 0.867$) from nearly independent data sources validates both signals, though some models diverge between the two evaluation modes.

5 Discussion

5.1 Formatting Bias Is Real but Bounded

Our results establish that markdown formatting independently influences user preferences in the Compar:IA arena. Each standard deviation of bold, lists, or headers increases win odds by 16–19%. This is a nontrivial effect at the individual battle level.

However, the aggregate impact on rankings is moderate. The overall correlation between standard and style-controlled ratings is 0.976—formatting bias reshuffles some models' positions but does not fundamentally reorder the leaderboard. The signal is dominated by genuine quality differences between models.

This finding is consistent with the LMSYS analysis of English-language data, which also found modest formatting effects. The cross-linguistic agreement suggests that formatting preferences are not culturally specific to English-speaking users—French users exhibit comparable biases.

5.2 Implications for Arena Design

The systematic benefit to formatting-heavy models has practical implications for arena operators:

1. **Ranking interpretation.** Models like mistral-medium-3.1 (rank 8 $\rightarrow$ 30 after style control) and mistral-large-2512 (rank 3 $\rightarrow$ 23) may be overranked due to formatting rather than content quality. Arena leaderboards should consider publishing both standard and style-controlled rankings.

2. **Reasoning model evaluation.** The current arena format may systematically undervalue reasoning models, which sacrifice formatting for reasoning depth. Specialized evaluation tracks or formatting-agnostic presentation (e.g., rendering all responses in plain text) could address this.

3. **Model development incentives.** If arena rankings drive model development priorities, formatting bias creates perverse incentives: teams may optimize for visually appealing output rather than substantive quality.

5.3 Endogeneity: Confounder or Mediator?

A fundamental interpretive challenge for observational style control is endogeneity. Our style features are covariates, not experimental manipulations, and two competing causal models fit the data:

Confounder hypothesis (formatting as bias). Users are partially "fooled" by visual presentation. Formatting creates a halo effect that inflates win probability independent of content quality. Under this model, style control removes a bias, and style-controlled rankings better reflect true model quality.

Mediator hypothesis (formatting as quality signal). Better models produce better-structured output *because they are more capable*—they understand when to use headers, lists, and bold emphasis to organize complex information. Formatting mediates the relationship between model capability and user preference. Under this model, style control inadvertently removes legitimate quality signal.

We present three empirical tests that shed light on this question, though they cannot fully resolve it.

Test 1: Quality–formatting correlation. We computed each model's average formatting intensity (composite of bold, lists, and headers per response) and correlated it with standard BT ratings across all 90 models. The correlation is strong and positive: Pearson $r = 0.65$ ($p < 10^{-11}$), Spearman $\rho = 0.74$ ($p < 10^{-16}$). Higher-rated models produce substantially more formatted output. This is consistent with the mediator hypothesis—better models do format more—but it is also consistent with the confounder hypothesis if formatting inflates ratings.

After style control, the correlation between model ratings and formatting intensity drops from $r = 0.65$ to $r = 0.48$. Style control weakens but does not eliminate the quality–formatting association, suggesting that the relationship is partially genuine (mediation) and partially artifactual (confounding).

Test 2: Tier-stratified style effects. If formatting is purely a mediator of quality, its effect on win probability should be similar regardless of model tier—good formatting should help a top model as much as a bottom model. If formatting is a confounder (bias), its effect might differ by tier, potentially helping weaker models more. We split the 90 models into three tiers of approximately equal size (top, middle, bottom) by standard BT rating and ran the style-controlled BT model separately on within-tier battles (Table 6, Figure 6).

Table 6: Style effects by battle pair tier (interaction model, implied total odds change per SD).

Feature	Bottom-bottom	Middle-middle	Top-top
Bold	+19.4%	+17.5%	+10.3%
Lists	+21.9%	+12.7%	+11.4%
Headers	+16.8%	+17.4%	+6.3%
N battles	20,180	12,824	7,796

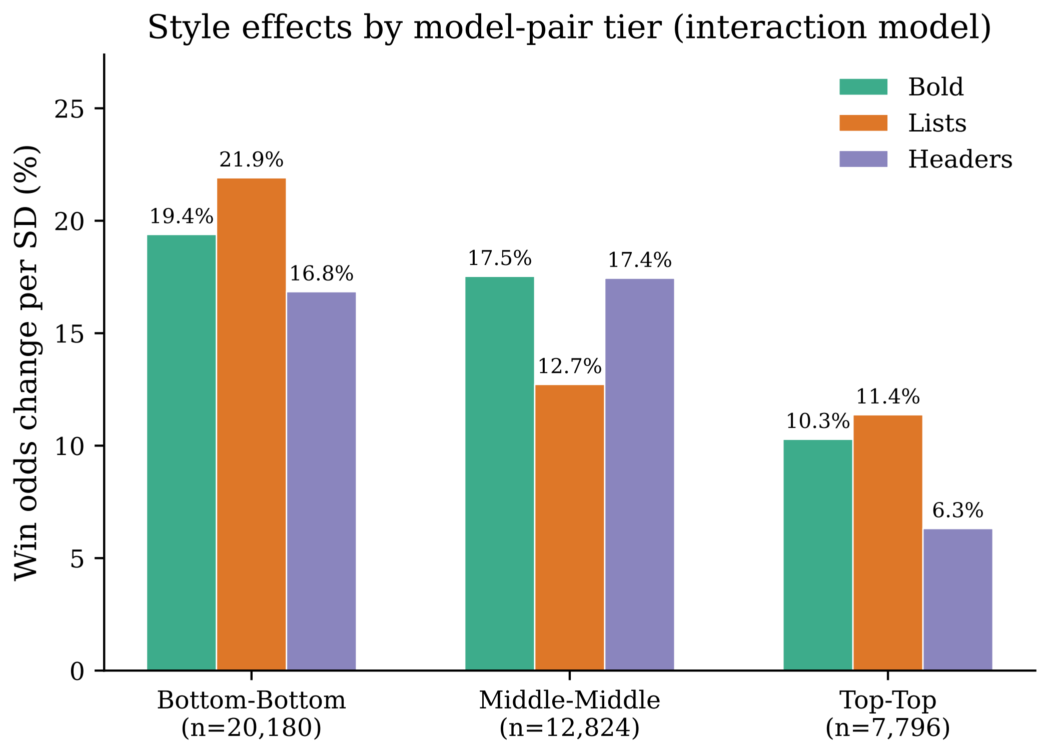

Figure 6: Style effects by model-pair tier. Formatting bias is roughly twice as large in bottom-tier battles as in top-tier, consistent with the confounder interpretation.

The style effect is roughly twice as large in bottom-tier battles as in top-tier battles for bold (+19.4% vs. +10.3%) and lists (+21.9% vs. +11.4%). Headers show a similar pattern (+16.8% vs. +6.3%). To test statistical significance, we fit an interaction model with tier $\times$ style terms in a unified logistic regression controlling for model identity. Of the six interaction terms (3 features $\times$ 2 tiers), three—top$\times$bold (bootstrap 95% CI: $[-0.16, -0.01]$), top$\times$lists ($[-0.17, -0.01]$), and top$\times$headers ($[-0.16, -0.02]$)—were statistically significant, all with negative signs indicating reduced style effects for top-tier battles. The three middle-tier interactions showed consistent negative signs but did not reach significance at the 5% level.

This gradient is more consistent with the confounder interpretation: formatting bias has a larger effect on user preferences when comparing weaker models, where content quality differences may be smaller and superficial presentation cues more decisive. Among top models—where content quality is consistently high—formatting matters less.

Test 3: Rating change as a function of formatting intensity. Across all 90 models, the rating change after style control (controlled rating minus standard rating) correlates at $r = -0.93$ ($p < 10^{-39}$) with formatting intensity (Figure 7). Models that format heavily lose the most rating points. This mechanical relationship confirms that style control operates as intended, but it does not resolve whether the removed signal was bias or quality.

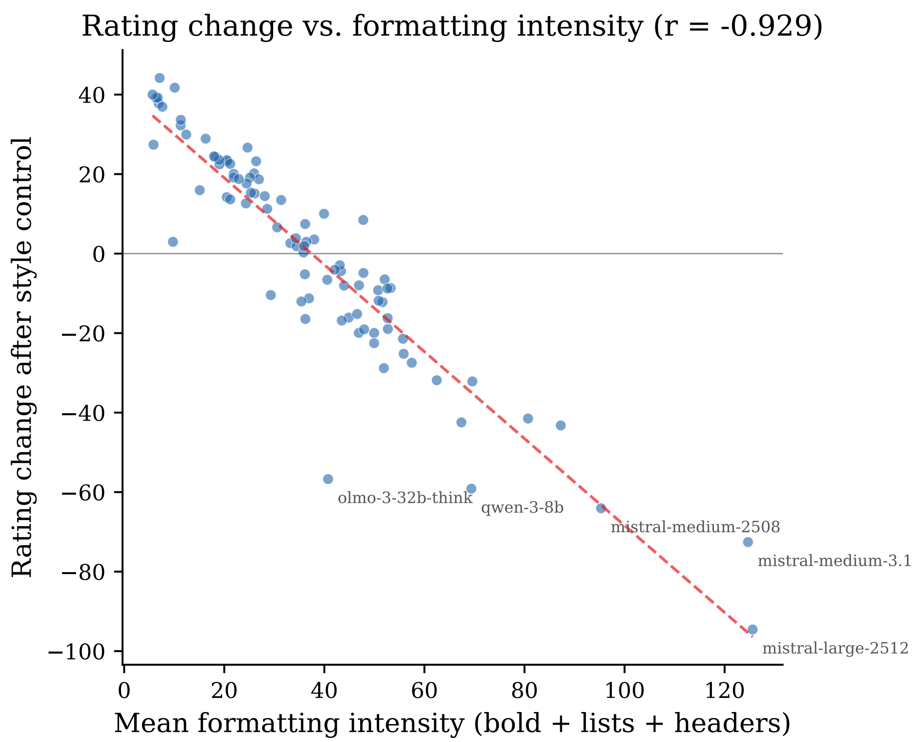

Figure 7: Rating change vs. formatting intensity ($r = -0.928$). Formatting-heavy models (right) lose the most rating points after style control.

Synthesis. The evidence suggests that formatting is *both* a partial mediator and a partial confounder—a common outcome in observational studies. Better models genuinely produce better-structured output (mediator component: $r = 0.48$ between controlled ratings and formatting), but formatting also exerts an independent influence on user preferences beyond what model quality would predict (confounder component: the tier gradient, where formatting effects are largest among weaker models). This dual role means that neither standard rankings (which include formatting bias) nor style-controlled rankings (which may overcorrect) are a definitive measure of model quality. We recommend that arena operators report both, allowing consumers to triangulate.

5.4 Qualitative Analysis of Winner-Flipping Battles

To move beyond aggregate statistics, we examine individual conversations where style control changes the predicted outcome. We define a "winner-flipping" battle as one where the standard BT model and the style-controlled BT model disagree on which model is stronger—i.e., model A has a higher standard rating but model B has a higher controlled rating, or vice versa.

Prevalence. Of 94,047 non-tie battles, 5,559 (5.9%) are winner-flipping.

Formatting asymmetry in flips. In flipped battles, the vote winner uses more total formatting (headers + lists + bold) in 53.3% of cases, while the vote loser formats more in 38.3% (Figure 8). This asymmetry—modest but consistent—suggests that formatting provides a marginal advantage in closely contested battles.

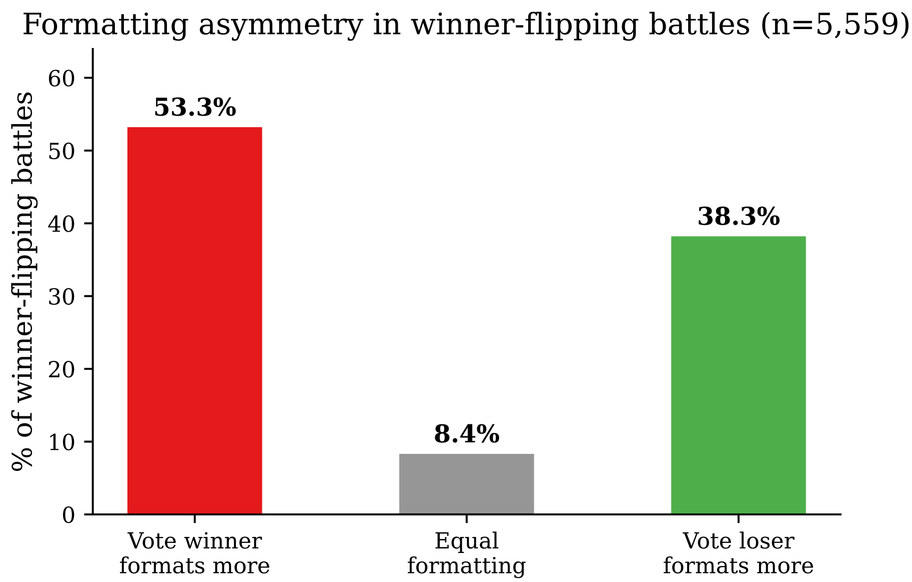

Figure 8: Formatting asymmetry in winner-flipping battles. The vote winner formats more in 53.3% of cases vs. 38.3% for the loser.

Which models flip? The models appearing most frequently in flips are a mix of heavy formatters whose standard ratings are inflated (mistral-large-2512: 642 flips, claude-4-5-sonnet: 448, o4-mini: 448) and lower-ranked models that serve as frequent opponents (ministral-8b-instruct-2410: 651, llama-3.1-405b: 543). This is expected: flips require two models whose relative ranking reverses after style control, which is most likely when one is a heavy formatter whose inflation creates a spurious advantage.

Illustrative examples. Manual inspection of 20 high-style-boost flips reveals three recurring patterns:

Pattern 1—Similar content, different packaging. In the most common pattern, both models provide substantively equivalent information, but the vote winner deploys dramatically more formatting. For instance, when asked to list famous athletes known for injuries, the heavily formatted version (41,474 chars; 103 headers, 412 list items, 652 bold spans) won over a concise version (15,388 chars; no headers, 34 list items, 36 bold spans) covering the same athletes.

Pattern 2—Appropriate brevity penalized. When asked to “describe BS1” (an ambiguous acronym), one model appropriately asked for clarification in 553 characters, while another produced a 33,372-character encyclopedic survey with 58 headers, 159 list items, and 487 bold spans. The user voted for the longer, formatted response despite the arguably more appropriate behavior of recognizing ambiguity.

Pattern 3—Users sometimes see through formatting. In roughly one-third of flips, the vote goes *against* the more formatted model. For example, when asked whether a French sentence is grammatically correct, a user preferred a concise 6,035-character answer over a 22,170-character analysis with 48 headers that dramatically overanalyzed a simple question. These counter-examples suggest that formatting bias, while real, is not deterministic.

User-reported formatting quality. The Compar:IA platform asks users to tag responses as having "clear formatting." Among winner-flipping battles with this attribute, the vote winner is tagged as having clear formatting in 17.7% of cases, versus only 2.4% for the vote loser. This 7:1 ratio suggests that when users explicitly evaluate formatting quality, they align with their vote—consistent with formatting acting as a conscious preference rather than a purely unconscious bias.

5.5 Limitations

Correlated features. Bold, lists, and headers are correlated (models that use one tend to use all three). The joint model partitions their shared effect, but the individual contributions may not be identifiable.

Response length excluded. We excluded response length from the style control to avoid removing genuine quality signal. This means our "style-controlled" rankings still contain any formatting-correlated length bias. Including length would likely increase the measured style effect and produce larger rank changes.

Bootstrap convergence. We used 1,000 bootstrap iterations for both style coefficients and BT ratings, providing stable confidence intervals suitable for publication.

Reaction-derived data. The conversion of binary like/dislike reactions to pairwise preferences produces structural tie inflation (46.8% vs. 30.7%), and the inner merge drops 45.4% of conversations with one-sided reactions. These are documented but not corrected.

Platform-specific population. Compar:IA users are predominantly French civil servants and technology-interested citizens. Results may not generalize to other populations.

Reasoning-only content filter. The exclusion of 25,778 battles (15.1%) where model response content was missing but reasoning content was present represents a substantial data loss. These battles are concentrated among specific models (e.g., gemini-3-pro-preview, gpt-5-mini, qwen3-30b-a3b), which reduces statistical power for those models. The filter is necessary—without it, zero-valued style features from missing text would be conflated with genuinely unformatted output—but it means our analysis covers a subset of the platform's interactions.

Tier stratification is endogenous. The tier-based interaction analysis (Section 5.3) stratifies on the standard BT rating, which itself includes formatting effects. This means tier assignments partially reflect the quantity we are trying to analyze. A fully exogenous stratification (e.g., by model parameter count or by a third-party benchmark) would strengthen the test, though the large tier boundaries make this concern modest in practice.

6 Conclusion

We find that markdown formatting—specifically bold text, lists, and headers—independently influences user preferences in the French Compar:IA LLM arena, with each feature increasing win odds by 16–19% per standard deviation. After controlling for these formatting effects, 78 of 90 models (87%) show statistically significant rating changes after Benjamini-Hochberg correction (FDR < 0.05), demonstrating that formatting bias is pervasive rather than concentrated in a few models. Formatting-heavy models are most affected: mistral-medium-3.1 drops 22 ranks and mistral-large-2512 drops 20 ranks. Reasoning models show a modestly positive mean rank change (+2.6) after style control, though the pattern is mixed across individual models.

These findings replicate and extend the LMSYS style control methodology to a non-English arena for the first time, demonstrating that formatting bias in LLM evaluation is not an English-language phenomenon. The practical implication is that arena rankings should be interpreted with awareness that they partially reflect formatting preferences rather than pure content quality. A notable methodological finding is that excluding battles with missing response content (a data pipeline artifact affecting 15.1% of battles) substantially changes the estimated reasoning model benefit, highlighting the importance of careful data cleaning in arena analyses.

A Data Quality Summary

Filter	Rows Removed	% of Original
Same-model pairs (votes)	503	0.34%
No-choice votes	3,994	2.68%
Duplicate conversation_pair_ids	3,401	2.28%
Same-model pairs (reactions)	352	0.39%
Even msg_index (user messages)	995	1.11%
Short responses (<10 chars)	431	0.48%
One-sided reactions (inner merge)	∼23,681 conv	45.4% of reaction conv
<think> contamination	98 conversations	0.07%
Reasoning-only content	25,778 battles	15.1% of combined

Table 7: Data quality filters applied. Final dataset: 145,347 battles across 90 models with ≥100 battles each.

B Top 20 Model Rankings

References

Benjamini, Yoav and Yosef Hochberg (1995). "Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing". In: *Journal of the Royal Statistical Society: Series B (Methodological)* 57.1, pp. 289–300. DOI: `10.1111/j.2517-6161.1995.tb02031.x`.

Std Rank	Model	Std Rating	Ctrl Rating	Ctrl Rank	ΔRank
1	gemini-3-pro-preview	1288.0	1285.0	1	0
2	gemini-3-flash-preview	1201.3	1196.1	2	0
3	mistral-large-2512	1176.9	1082.4	23	−20
4	mistral-medium-2508	1169.7	1105.6	9	−5
5	gemini-2.5-flash	1166.8	1137.9	3	+2
6	magistral-medium	1147.9	1128.8	4	+2
7	qwen3-max-2025-09-23	1145.1	1119.9	6	+1
8	mistral-medium-3.1	1131.7	1059.1	30	−22
9	gemini-2.0-flash	1129.7	1121.0	5	+4
10	gpt-5.1	1126.2	1103.7	10	0
11	deepseek-v3-0324	1116.9	1100.7	12	−1
12	glm-4.6	1114.5	1087.1	21	−9
13	deepseek-r1-0528	1112.4	1092.5	16	−3
14	gemma-3-27b	1111.2	1102.4	11	+3
15	claude-4-5-sonnet	1110.8	1089.3	20	−5
16	deepseek-chat-v3.1	1109.7	1090.7	19	−3
17	gpt-5.2	1108.4	1091.9	17	0
18	grok-4.1-fast	1102.8	1105.7	8	+10
19	DeepSeek-V3.2	1099.8	1093.3	15	+4
20	deepseek-v3-chat	1099.6	1091.5	18	+2

Table 8: Top 20 models by standard BT rating. Full 90-model table available in supplementary materials.

Bradley, Ralph Allan and Milton E. Terry (1952). "Rank analysis of incomplete block designs: I. The method of paired comparisons". In: *Biometrika* 39.3–4, pp. 324–345. DOI: `10.1093/biomet/39.3-4.324`.

Chiang, Wei-Lin et al. (2024). "Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference". In: *Proceedings of the 41st International Conference on Machine Learning*. arXiv:2403.04132, pp. 8359–8388. URL: `https://proceedings.mlr.press/v235/chiang24b.html`.

Li, Tianle et al. (2024). "From Crowdsourced Data to High-Quality Benchmarks: Arena Hard and BenchBuilder Pipeline". In: *arXiv preprint arXiv:2406.11939*. DOI: `10.48550/arXiv.2406.11939`. URL: `https://arxiv.org/abs/2406.11939`.

Termignon, Lucie et al. (2026). "compar:IA: The French Government's LLM Arena to Collect French-Language Human Prompts and Preference Data". In: Working paper.

Zheng, Lianmin et al. (2023). "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena". In: *Advances in Neural Information Processing Systems* 36, pp. 46595–46623. URL: `https://proceedings.neurips.cc/paper_files/paper/2023/hash/91f18a1287b398d378ef22505bf41832-Abstract-Datasets_and_Benchmarks.html`.